

```
In [1]: import cv2
import re
from matplotlib import pyplot as plt
import numpy as np
import pytesseract
import pymongo
from PIL import Image
im_file = "C:/Users/neeta/temp/raducky_aadhaar.jpg"
img = cv2.imread(im_file)
```

```
In [2]: def display(im_path):
    dpi = 80
    im_data = plt.imread(im_path)
    height, width = im_data.shape[:2]
    figsize = width / float(dpi), height / float(dpi)
    fig = plt.figure(figsize=figsize)
    ax = fig.add_axes([0, 0, 1, 1])
    ax.axis('off')
    ax.imshow(im_data, cmap='gray')
    plt.show()
display(im_file)
```



```
In [3]: def grayscale(image):
    return cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
In [4]: gray_image = grayscale(img)
cv2.imwrite("C:/Users/neeta/temp/graya1.jpg", gray_image)
```

Out[4]: True

```
In [5]: display("C:/Users/neeta/temp/graya1.jpg")
```



```
In [1]: def getSkewAngle(cvImage) -> float:
newImage = cvImage.copy()
blur = cv2.GaussianBlur(newImage, (9, 9), 0)
thresh = cv2.threshold(blur, 0, 255, cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU)[1]

kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (30, 5))
dilate = cv2.dilate(thresh, kernel, iterations=5)

contours, hierarchy = cv2.findContours(dilate, cv2.RETR_LIST, cv2.CHAIN_APPROX_
contours = sorted(contours, key = cv2.contourArea, reverse = True)

largestContour = contours[0]
minAreaRect = cv2.minAreaRect(largestContour)

angle = minAreaRect[-1]
if angle < -45:
    angle = 90 + angle
return -4.023 * angle
```

```
In [2]: def rotateImage(cvImage, angle: float):
newImage = cvImage.copy()
(h, w) = newImage.shape[:2]
center = (w // 2, h // 2)
M = cv2.getRotationMatrix2D(center, angle, 1.0)
newImage = cv2.warpAffine(newImage, M, (w, h), flags=cv2.INTER_CUBIC, borderMod
return newImage

def deskew(cvImage):
angle = getSkewAngle(cvImage)
return rotateImage(cvImage, -1.0 * angle)
```

```
In [8]: fixim = deskew(gray_image)
cv2.imwrite("C:/Users/neeta/temp/rotfix.jpg", fixim)
display("C:/Users/neeta/temp/rotfix.jpg")
```



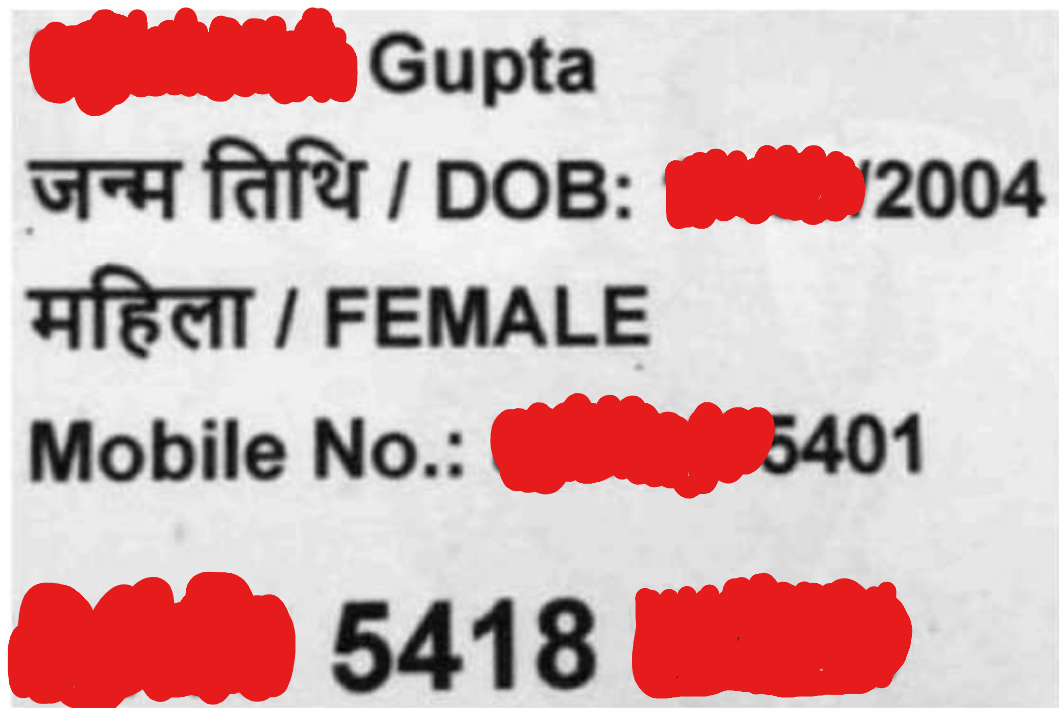
```
In [9]: print(img.shape)
```

```
(626, 1020, 3)
```

```
In [10]: cropped_image = fixim[210:505, 340:785]  
cv2.imwrite("C:/Users/neeta/temp/CropAd.jpg", cropped_image)
```

```
Out[10]: True
```

```
In [11]: display("C:/Users/neeta/temp/CropAd.jpg")
```



```
In [12]: thresh, im_bw = cv2.threshold(cropped_image, 140, 190, cv2.THRESH_BINARY)  
cv2.imwrite("C:/Users/neeta/temp/bww.jpg", im_bw)
```

Out[12]: True

In [13]: `display("C:/Users/neeta/temp/bww.jpg")`

[REDACTED] Gupta
जन्म तिथि / DOB: [REDACTED]/2004
महिला / FEMALE
Mobile No.: [REDACTED] 5401
[REDACTED] 5418 [REDACTED]

In [14]: `im_file = "C:/Users/neeta/temp/bww.jpg"`

In [15]: `img = Image.open(im_file)`
`ocr_resultf = pytesseract.image_to_string(img)`

In [16]: `print(ocr_resultf)`

[REDACTED] Gupta
wa fafer / DOB: [REDACTED]/2004

after) FEMALE

Mobile No.: [REDACTED] 5401
[REDACTED] 5418 [REDACTED]

In [78]: `pattern = re.compile('\d.*\d')`
`matches = pattern.finditer(ocr_resultf)`

In [79]: `pattern2 = re.compile('\D.*\D')`
`matchesnam = pattern2.finditer(ocr_resultf)`

In [80]: `pattern3 = re.compile('(MALE|..MALE)\s')`
`matchesgen = pattern3.finditer(ocr_resultf)`

In [81]: `for match in matches:`
`print(match)`

`<re.Match object; span=(30, 40), match='[REDACTED]/2004'>`
`<re.Match object; span=(69, 79), match='[REDACTED] 5401'>`
`<re.Match object; span=(80, 94), match='[REDACTED] 5418 [REDACTED]'>`

```
In [82]: for matche in matchesnam:
        print(matche)
```

```
<re.Match object; span=(0, 14), match='Gupta\n'>
<re.Match object; span=(14, 41), match='wa fafer / DOB: 2004\n'>
<re.Match object; span=(41, 56), match='\nafter) FEMALE\n'>
<re.Match object; span=(56, 80), match='\nMobile No.: 5401\n'>
<re.Match object; span=(84, 95), match=' 5418 \n'>
```

```
In [83]: for matchee in matchesgen:
        print(matchee)
```

```
<re.Match object; span=(49, 56), match='FEMALE\n'>
```

```
In [95]: f1 = ocr_resultf[0:13]
        f2 = ocr_resultf[30:40]
        f3 = ocr_resultf[49:55]
        f4 = ocr_resultf[69:79]
        f5 = ocr_resultf[80:94]
```

```
In [96]: print(f1)
        print(f2)
        print(f3)
        print(f4)
        print(f5)
```

```
Gupta
/2004
FEMALE
5401
5418
```

```
In [97]: if __name__ == "__main__":
        client = pymongo.MongoClient("mongodb://localhost:27017")
        print(client)
        print(client.list_database_names())
```

```
MongoClient(host=['localhost:27017'], document_class=dict, tz_aware=False, connect=True)
['OCR', 'admin', 'config', 'local', 'sample']
```

```
In [98]: db = client['OCR']
        collection = db['Aadhaar']
```

```
In [101... dic1 = {'Name': f1, 'D.O.B.' : f2, "Sex" : f3, 'Mobile' : f4, 'Aadhaar Number' :
        print(dic1)
```

```
{'Name': 'Gupta', 'D.O.B.': '2004', 'Sex': 'FEMALE', 'Mobile': '5401', 'Aadhaar Number': '5418'}
```